



Lab Session

Week 11: Dictionaries - Key-Value Magic

7th Grade Computer Science - Module 2

Name: _____ Partner: _____

Lab Goals

By the end of this lab, you will be able to:

- Create dictionaries and access values using keys
- Use dictionary methods like `.get()`, `.keys()`, `.values()`, and `.items()`
- Apply the frequency counter pattern to count occurrences
- Choose between lists and dictionaries for different problems

Getting Started

1. Open Thonny on your computer
2. Create a new file called `mod2.week11.lab.py`
3. Save it in your Module2 folder
4. Type your name and your partner's name as comments at the top

 **Pair Programming:** Switch who types every 5 minutes!

1 Exercise 1: Your First Dictionary (8 minutes)

Let's create a dictionary to store information about your favorite cricket players and their scores.

```
1 # Create a dictionary of cricket player scores
2 players = {"Babar": 85, "Rizwan": 72, "Fakhar": 63}
3
4 # Print the entire dictionary
5 print(players)
6
7 # Access Babar's score
8 print(f"Babar's score: {players['Babar']}")
```

Your Turn: Add to the code above:

1. Add two more players to the dictionary:

```
1 players[_____] = _____
2 players[_____] = _____
```

2. Update Rizwan's score to 91:

```
1 _____
```

3. Print how many players are in your dictionary:

```
1 print(f"Total players: {_____}")
```

✔ **Checkpoint:** Show your teacher your dictionary with 5 players!

2 Exercise 2: Safe Dictionary Access (8 minutes)

What happens when you try to access a key that doesn't exist? Let's find out!

```
1 grades = {"Ali": 85, "Sara": 92, "Ahmed": 78}
2
3 # This will crash! (Don't run yet - predict first)
4 # print(grades["Hassan"])
5
6 # Safe way using .get()
7 hassan_grade = grades.get("Hassan", 0)
8 print(f"Hassan's grade: {hassan_grade}") # Returns 0 (default)
9
10 # Check if key exists first
11 if "Hassan" in grades:
12     print(grades["Hassan"])
13 else:
14     print("Hassan not found in grades")
```

Your Turn: Complete this safe grade lookup function:

```
1 def lookup_grade(student_name):
2     grades = {"Ali": 85, "Sara": 92, "Ahmed": 78}
3
4     # Use .get() to safely return grade or "Not enrolled"
5     return grades.get(_____, _____)
6
7 # Test your function
8 print(lookup_grade("Ali")) # Should print: 85
9 print(lookup_grade("Hassan")) # Should print: Not enrolled
```

💡 **Tip:** Always use `.get(key, default)` when you're not sure if a key exists!

3 Exercise 3: Dictionary Methods (10 minutes)

Dictionaries have powerful methods to access keys, values, and both together.

```
1 # Shop inventory
2 inventory = {"samosas": 25, "pakoras": 40, "chai": 100}
3
4 # Get all product names (keys)
5 products = list(inventory.keys())
6 print(f"Products: {products}")
7
8 # Get all quantities (values)
9 quantities = list(inventory.values())
10 print(f"Quantities: {quantities}")
```

```

11
12 # Get everything as pairs
13 all_items = list(inventory.items())
14 print(f"All items: {all_items}")

```

Your Turn: Use dictionary methods to solve these:

```

1 inventory = {"samosas": 25, "pakoras": 40, "chai": 100, "jalebis": 15}
2
3 # 1. Calculate total items in stock (sum of all values)
4 total_stock = sum(_____)
5 print(f"Total items: {total_stock}")
6
7 # 2. Print each item and its quantity using .items()
8 print("\n--- Inventory Report ---")
9 for _____, _____ in inventory.items():
10     print(f"{_____}: {_____} units")
11
12 # 3. Find the product with the most stock
13 max_product = ""
14 max_quantity = 0
15 for product, quantity in inventory.items():
16     if _____:
17         max_product = product
18         max_quantity = quantity
19 print(f"\nMost stocked: {max_product} ({max_quantity} units)")

```

✔ **Checkpoint:** Can you explain what `.keys()`, `.values()`, and `.items()` return?

4 Exercise 4: Frequency Counter Pattern (12 minutes)

One of the most useful dictionary patterns is counting how often things appear!

```

1 # Count letters in a word
2 word = "mississippi"
3 letter_count = {}
4
5 for letter in word:
6     letter_count[letter] = letter_count.get(letter, 0) + 1
7
8 print(letter_count) # {'m': 1, 'i': 4, 's': 4, 'p': 2}

```

Your Turn: Use the frequency counter pattern for these challenges:

Challenge A: Count word frequency in a sentence

```

1 sentence = "the cat sat on the mat the cat was fat"
2 words = sentence.split() # Splits into list of words
3
4 word_count = {}
5 for word in words:
6     # Complete the frequency counter pattern:
7     _____
8
9 print(word_count)
10 # Expected: {'the': 3, 'cat': 2, 'sat': 1, 'on': 1, ...}
11

```

```

12 # Which word appears most often?
13 most_common = ""
14 highest_count = 0
15 for _____, _____ in word_count.items():
16     if _____:
17         most_common = _____
18         highest_count = _____
19
20 print(f"Most common word: '{most_common}' ({highest_count} times)")

```

Challenge B: Count dice roll results

```

1 import random
2
3 # Simulate 100 dice rolls
4 rolls = [random.randint(1, 6) for _ in range(100)]
5
6 # Count how many times each number appeared
7 roll_count = {}
8 for roll in rolls:
9     -----
10
11 # Print results
12 print("Dice Roll Results:")
13 for number in sorted(roll_count.keys()):
14     print(f" {number}: {roll_count[number]} times")

```

 **Pair Programming:** Switch who types every 5 minutes!

5 Exercise 5: Lists vs Dictionaries (8 minutes)

When should you use a list? When should you use a dictionary? Let's compare!

Scenario: Store student grades and look them up by name.

Using Lists (the hard way):

```

1 names = ["Ali", "Sara", "Ahmed"]
2 grades = [85, 92, 78]
3
4 # Find Sara's grade
5 def find_grade_list(name):
6     for i in range(len(names)):
7         if names[i] == name:
8             return grades[i]
9     return "Not found"
10
11 print(find_grade_list("Sara"))

```

Using Dictionary (the easy way):

```

1 grades = {"Ali": 85,
2           "Sara": 92,
3           "Ahmed": 78}
4
5 # Find Sara's grade
6 def find_grade_dict(name):
7     return grades.get(name,
8                       "Not found")
9
10 print(find_grade_dict("Sara"))

```

Your Turn: For each scenario, write **List** or **Dictionary**:

Scenario	Best Choice
Storing the top 10 high scores in order	_____
Looking up a student's ID number by their name	_____
Keeping track of items in a shopping cart (in order added)	_____
Counting how many of each fruit you have in stock	_____
Storing the steps in a recipe (order matters!)	_____
Mapping country names to their capitals	_____

6 Exercise 6: Build a Contact Manager (12 minutes)

Let's put everything together! Build a simple contact manager program.

```

1  # Contact Manager
2  contacts = {}
3
4  def add_contact(name, number):
5      """Add a new contact to the dictionary"""
6      contacts[_____] = _____
7      print(f"Added {name}!")
8
9  def find_contact(name):
10     """Find a contact by name, return number or 'Not found'"""
11     return contacts.get(_____, _____)
12
13  def delete_contact(name):
14     """Delete a contact if they exist"""
15     if _____ in contacts:
16         del contacts[_____]
17         print(f"Deleted {name}")
18     else:
19         print(f"{name} not found!")
20
21  def show_all_contacts():
22     """Display all contacts"""
23     if len(contacts) == 0:
24         print("No contacts saved!")
25     else:
26         print("\n--- Contact List ---")
27         for _____, _____ in contacts._____.():
28             print(f" {_____}: {_____}")
29
30  # Test your contact manager!
31  add_contact("Mom", "0300-1111111")
32  add_contact("Dad", "0300-2222222")
33  add_contact("Best Friend", "0321-3333333")
34
35  show_all_contacts()
36
37  print(f"\nMom's number: {find_contact('Mom')}")
38  print(f"Teacher's number: {find_contact('Teacher')}")
39
40  delete_contact("Best Friend")
41  show_all_contacts()

```

✔ **Checkpoint:** Does your contact manager add, find, delete, and show all contacts correctly?

🏆 Challenge Yourself

Extension: Add a menu system with a while loop so users can interactively manage contacts!

```
1 while True:
2     print("\n1. Add Contact  2. Find Contact  3. Show All  4. Quit")
3     choice = input("Choose: ")
4     # Add your code to handle each choice...
```

7 Lab Summary

✅ **Checkpoint:** Final checkpoint - make sure you have:

- ☐ Created dictionaries with key-value pairs
- ☐ Used `.get()` for safe access with defaults
- ☐ Used `.keys()`, `.values()`, and `.items()` methods
- ☐ Applied the frequency counter pattern
- ☐ Compared when to use lists vs dictionaries
- ☐ Built a working contact manager

Reflection (3 minutes)

Rate your understanding of today's concepts:

Concept	🔴 Need Help	🟡 Getting It	🟢 Got It!
Creating and accessing dictionaries	☐	☐	☐
Using <code>.get()</code> for safe access	☐	☐	☐
Frequency counter pattern	☐	☐	☐
Choosing lists vs dictionaries	☐	☐	☐

Which data structure would you use for:

1. Storing student grades by name: _____
2. Keeping a ranked list of top players: _____
3. Counting word frequency in text: _____

One thing that clicked today: _____

One thing to practice more: _____

🔑 **Key-Value Master! Great work today!**